



AMTEC

AGRICULTURAL MACHINERY TESTING AND EVALUATION CENTER

TEST REPORT NO. 2015 - 99



NEW TECH WALK-BEHIND RICE COMBINE HARVESTER Model : GY4L-0.9B

NEW TECH GY4L-0.9B Walk-Behind Rice Combine Harvester

Test Report No.	:	2015 - 99
Date of Test	:	June 2, 2015
Test Valid Until	:	June 2, 2020
Brand	:	NEW TECH
Model	:	GY4L-0.9B
Type	:	Walk-behind, crawler-type Rice combine harvester
Manufacturer	:	SICHUAN GANGYI TECHNOLOGY GROUP CO., LTD. Chengdu, Sichuan, China
Test Requested by	:	ASIAN GULF DISTRIBUTOR 496 Old Samson Road, EDSA, Caloocan City
Machine Classification	:	Commercial

SUMMARY

The NEW TECH GY4L-0.9B Rice Combine Harvester was tested for field Performance. It was tested based on original settings set by the manufacturer.

The NEW TECH GY4L-0.9B Rice Combine Harvester was tested using N20 rice variety with a Grain Moisture Content of 24.8 %, wet basis, and average Plant Height of 846 mm. Based on the results of the test, the NEW TECH GY4L-0.9B Rice Combine Harvester had an Actual Field Capacity of 0.081 ha/h or 12.3 h/ha and Field Efficiency of 58.9% at an average Travelling Speed of 1.38 kph. It had a Threshed Grain Output Capacity of 386.3 kg/h with threshed grain Purity of 97.2 %. The rice combine harvester had a Total Grain Loss of 0.85 % consisting of 0.04 % Header Loss, 0.81 % Separation/Blower Loss, no Uncut Loss and no Unthreshed Loss.

Grain quality analysis of threshed grains showed a net percent Cracked Grain of 3.67 % and no Broken Grains.

The Fuel Consumption Rate of the rice combine's diesel engine was 1.02 L/h.

During the test, one person was needed to operate the rice combine, for operating the machine and for bagging of the threshed grain output.

No breakdown or failure of components occurred during the test.

PURPOSE AND SCOPE OF TEST

The machine was tested for verification of its performance. The test was conducted at a farmer's field in Barangay Malainen Bago, Naic, Cavite and was evaluated on the following items:

1. Field Capacity
2. Field Efficiency
3. Output Capacity
4. Grain losses
 - a. Header Loss
 - b. Separation/Blower Loss
 - c. Uncut Loss
 - d. Unthreshed Loss
5. Fuel Consumption

METHOD OF TEST

The Rice Combine Harvester was tested based on "AMTEC Methods of Test for Rice Combine Harvesters".

OPERATOR'S MANUAL

No manual was submitted.

DESCRIPTION OF THE MACHINE

The NEW TECH GY4L-0.9B Walk-behind Rice Combine Harvester (**Figure 1**), was a walk-behind, crawler-type harvesting machine which combined the cutting and threshing operations in one machine. It was powered by an 8.96 kW diesel engine which drove the transmission systems of the machine. The rice combine consisted of reel assembly, cutting assembly, screw and chain conveyors, threshing assembly, traveling device and control levers.

The reel assembly (**Figure 2**) with its rotary motion guides the rice plant to the cutting assembly. It consisted of five anchor bars, with each bar having 14 pieces of protruding prongs. The reel assembly is not adjustable.

The machine's cutting device consisted of reciprocating cutting knives (**Figure 3**). It had a cutting width of 1.0 m. The height of cut was adjustable.

The rice combine's delivery device to its threshing unit consisted of a screw conveyor (**Figure 4**) and a chain conveyor. The screw conveyor conveyed the cut plants to a chain conveyor which in turn delivered the cut plants to the threshing chamber of the combine.

The threshing assembly was an axial-flow type threshing mechanism. It consisted of an open-type threshing cylinder (**Figure 5**) provided with four anchor bars. Each anchor bar was welded with 10 peg teeth.

The cleaning assembly had a centrifugal fan which separated the chaffs, straws, empty grains and other light impurities from the threshed grain. It also had front and rear sieving plates.

The rice combine was provided with output grain sack support (**Figure 6**) for bagging of threshed grain.

The machine had a crawler-type (**Figure 7**) traveling device made of rubber.

The rice combine harvester was provided with control levers (**Figure 8**) at left, center, and right handles accessible to the operator.



Figure 1. NEW TECH Walk-behind
Rice Combine Harvester
Model: GY4L-0.9B



Figure 2. Reel Assembly



Figure 3. Reciprocating cutting knife



Figure 4. Screw conveyor



Figure 5. Threshing cylinder

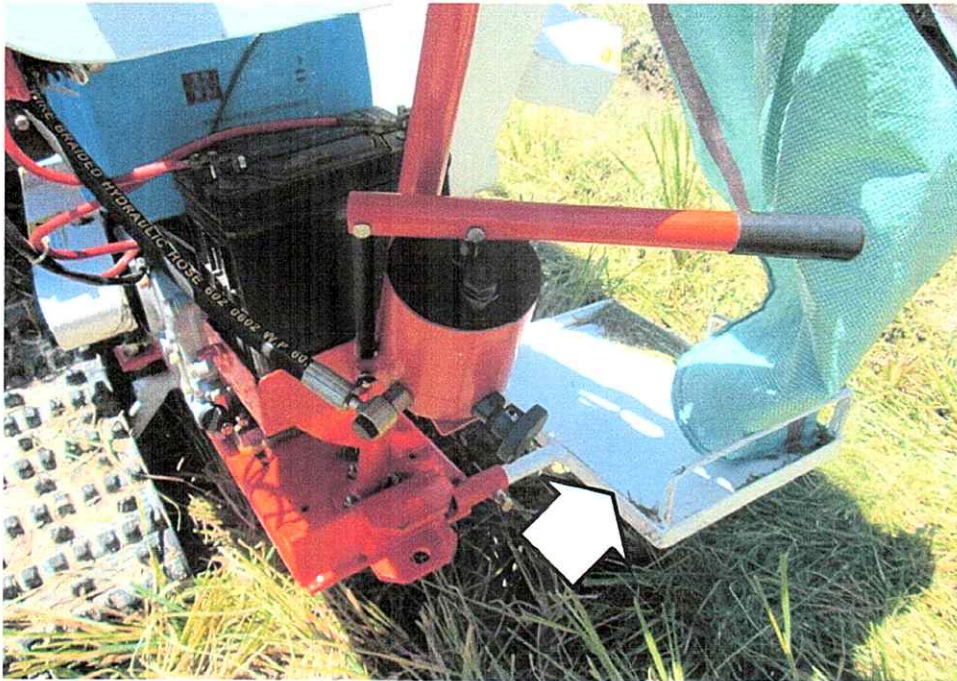


Figure 6. Output grain sack support



Figure 7. Crawler type travelling device



Figure 8. Control levers

RESULTS AND DISCUSSIONS

Results of field performance tests are shown in Table 1.

Performance

The performance of the rice combine was evaluated in terms of its field capacity, field efficiency, output capacity, grain losses and grain quality.

Actual Field Capacity. It is the ratio of the area harvested and the total time of operation. The Field Capacity of the machine was 0.081 ha/h or 12.3 h/ha.

Field Efficiency. It is the ratio of the Actual Field Capacity and the Theoretical Field Capacity, expressed in percent. The Field Efficiency of the machine was 58.9%.

Header Loss. This refers to the grains that have fallen to the ground due to machine's cutting operation. Header Loss was 0.04%.

Separation/Blower Loss. This refers to the grains that come out of the threshing chamber with the straw which is also the outlet of the blower. Separation/Blower Loss was 0.81 %.

Uncut Loss. This refers to the panicles that were not cut or missed by the reaping mechanism of the machine. There was no Uncut Loss.

Unthreshed Loss. This refers to grains that remained in the panicles of the plant fed into the threshing chamber. There was no Unthreshed Loss.

Grain quality. Quality of threshed grain was determined in terms of percent cracked and broken grains. Results showed that the threshed output had a net percentage Cracked Grains of 3.67 % and no Broken Grains. (Cracked grains refer to those grains whose endosperms are partially fractured while broken grains refer to those grains that are split into 2 or more pieces.)

Fuel Consumption

The rice combine's 8.96 kW diesel engine had a Fuel Consumption Rate of 1.02 L/h.

Labor Requirement

During testing, one person was needed to operate the rice combine (**Figure 9**), for operating the machine and for bagging of threshed grain.

Observations

1. The rice combine was easy to maneuver at corners (**Figure 10**).
2. Operation of the rice combine was stable (**Figure 11**).



Figure 9. Operator of the rice combine



Figure 10. Turning at corners



Figure 11. The stable operation of the rice combine

FIELD PERFORMANCE TEST

Machine tested : NEW TECH Rice Combine Harvester
Model : GY4L-0.9B

Location of test : Barangay Malainen Bago, Naic, Cavite

Date of test : June 2, 2015

Table 1. Results of field performance test

ITEMS		TEST DATA
A.	Condition of Crop	
1.	Variety	: N20
2.	Mode of Planting	: Broadcast
3.	Plant Height (mm)	: 846
4.	Moisture Content of Grains (%)	: 24.8
B.	Field Conditions	
1.	Type of Soil	: No data available
2.	Soil Resistance (kg/cm ²)	: 1.71

Table 1. Results of field performance test (cont'n.)

ITEMS		TEST DATA
C.	Field Performance	
1.	Method of Operation	: Circuitous
2.	Travelling Speed (kph)	: 1.38
3.	Effective Cutting Width (mm)	: 720
4.	Height of Cut or Stubbles (mm)	: 383
5.	Actual Field Capacity	
	a) ha/h	: 0.081
	b) h/ha	: 12.3
6.	Theoretical Field Capacity	
	a) ha/h	: 0.138
	b) h/ha	: 7.25
7.	Field efficiency (%)	: 58.9
8.	Threshed Grain Output Capacity (kg/h)	: 386.3
9.	Purity of threshed grain (%)	: 97.2
10.	Header Loss (%)	: 0.04
11.	Separation/Blower Loss (%)	: 0.81
12.	Unthreshed Loss (%)	: Nil

Test Engineers:


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24 AUGUST 2015
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N.B.

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